Signed Cylinder Propagation and Marked Polygons: Jordan-Adapted Word Tails and Stable Trace Windows

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Abstract

The assigned honeycomb manuscript passes from finite polynomial identities to a signed horizontal gas and extracts its cylinder coefficient without dividing by a possibly vanishing trace. We retain that mechanism and prove a local extension: a polynomial word-length moment of order equal to the largest peripheral Jordan degree suffices in place of exponential moments. An explicit discarded-word remainder separates complementary spectral decay from the required tail modulus. Anchored perturbation words with algebraic decay provide a concrete new sufficient regime. We extend the argument to the marked two-gap frequency grouping and give finite-window reconstruction bounds, including perturbations of both measured traces. These are conditional operator results, not an independent proof of the source cylinder or planar polygon exponents. Physical signs, selected-vector normalization, ordered limits, and geometric transfer remain essential. Original private finite-moment and physical-metric proofs are used only within their freshly read scope. Contribution assessment: C3/5, a source-specific weakening of word summability with quantitative cancellation-safe trace extraction, not a new polygon exponent or global proof certification.

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1 Model, status, and proof dependencies

1.1 Which law and which weight

The source [1] uses the honeycomb graph dual to the triangular tessellation of side one, with critical vertex weight

$$x = (2+\sqrt{2})^{-1/2}.$$

A first-exit walk between boundary ports visits distinct triangle centers; its weight is x^N for N visited centers. Closed polygons are unoriented and unrooted, their bond and vertex counts agree, and each translation class is counted once in the planar polygon sum. These conventions cannot be replaced by a stationary Markov law or by the uniform law on walks of a fixed length.

On the balanced cylinder there are $M = 2m$ sites, with m even, and two marked face centers separated by m sites along a horizontal cut. The partition $G_m(g)$ sums mutually vertex-disjoint simple cycles separating the marks, including the permitted noncontractible cycles, with cycle weight $gx^{|\lambda|}$ and empty configuration of weight one. Closed-rim height tends to infinity at fixed M. The source asserts

$$G_m(2) = m^{1/6+o(1)}, \qquad \mathcal{Z}(R) = R^{1/12+o(1)}, \qquad \mathbb{E}_R^{\mathrm{chord}} N = R^{4/3+o(1)}. \tag{1}$$

The last expectation is for ordered first-exit chords in a finite regular hexagon, both ports free, conditioned by the geometric restriction $\operatorname{diam}\gamma \geq R/10$ and normalized by their raw partition. The source also asserts a length-square weighted planar polygon bound $H^{2/3+o(1)}$ under its diameter cutoff. These are source conclusions, not conclusions established afresh by our finite checks.

The signed matrix and gas expressions are intermediate amplitudes. Positivity of $G_m(g)$ for real $g \geq 0$ or of a marked polygon sum follows from its combinatorial identification, not from positivity of each operator entry. In particular neither Perron–Frobenius reasoning nor orthogonal peripheral projections may be silently imposed on the horizontal operator.

1.2 Architecture and retained details

The source chain has the following distinct obligations.

1. The local parafermionic balance uses spin $3/8$, with $2x\cos((3/8)\pi/3) = 1$. Rhombic exchange, unitarity and fusion identities are finite algebraic statements. Fusion maps are not isometries: their Gram factors must be retained even though similarity in an invariant-space trace introduces no extra Gram scalar.

2. The finite inhomogeneous cylinder has a common fixed eigencharacter. A determinant construction gives its projector. Its full denominator has two factors, whereas the selected physical left and right vectors admit the particular smaller normalizer $N(X)$ proved in the source. One must not cancel the extra factor in every projector entry. The selected components, after the stated monomial normalization, have individual degree $M - 1 - \sigma^2$. The scalar $N(X)^2 G$ has individual

degree $2(M-1)$. Fusion and deletion fix the unbalanced interpolation problem; in the balanced case a remaining coefficient is fixed by the paired-scale limit.

3. The finite contour expansion terminates because repeated selection of an own-site residue repeats a Wick field and annihilates the Pfaffian. Gaussian contractions use bilinear, not Hermitian, covariance for complex arguments. Difference-channel neutrality is required before a singular zero-frequency limit. Fixed-current conditional oscillator traces are not by themselves permission to average an infinite expansion.

4. Pre- and post-collision stability, the physical charge quotient, and a confining gauge construct Hilbert–Schmidt horizontal operators. The memory space includes the residual interaction tail, rather than discarding it after a finite block. The collision residue creates the fused pair label. The physical pair and the half-normalized coordinate in a coercivity calculation must not be confused.

5. At fixed site lists the periodized trace difference has a limit. The signed coefficient lemma, not a quotient by one trace, extracts the central amplitude. A homogeneous Pfaffian normalization and an analytic edge-profile argument then supply the cylinder exponent and its lower bound.

6. Localization controls escaped cycles by an ordered incompatibility sum including self-pairs. Marked flips require a separate exact phase cancellation, unequal-list gas, shifted-height suppression at displacement $1/8$, mixed spectral estimate, and a two-gap coefficient lemma. Only thereafter does an injective large-period planar projection and a dyadic bond count yield the marked polygon estimate.

We preserve all six obligations. Sections 3–6 modify only the coefficient-extraction and defect-word interface. The source already handles Jordan blocks and cancellation; those features are not claimed here as new.

1.3 Current primary context

The primary Annals page for Duminil-Copin and Smirnov [2] identifies the connective constant $\sqrt{2+\sqrt{2}}$ and the parafermionic method. Its abstract distinguishes that theorem from the conjectural scaling-limit completion. Krachun and Panagiotis [3] state a high-probability $O(n/\log n)$ radius bound for length-n self-avoiding walks, with critical bridge decay as an ingredient. This is a different law and assertion from the source finite-chord mean in (1). These primary metadata and abstracts were freshly checked on October 9, 2026; their complete external proofs were not audited. No independent validation or exhaustive novelty search for the assigned manuscript is inferred from the public search.

2 The exact signed trace interface

All norms below are physical Hilbert-space norms. Let $\mathcal{H}$ be a complex Hilbert space, $E \in \mathcal{S}_2(\mathcal{H})$ have spectral radius one, and $C \in \mathcal{S}_1(\mathcal{H})$. Here $\mathcal{S}_2$ and $\mathcal{S}_1$ denote the Hilbert–Schmidt and trace-class ideals. Compactness gives a finite nonempty peripheral set $\mathcal{P} = \{\zeta \in \operatorname{spec} E : |\zeta| = 1\}$. Let P_ζ be its Riesz projections, $m_\zeta = \operatorname{rank} P_\zeta$, and

$$J_\zeta = (E - \zeta I)P_\zeta, \qquad q_\zeta \geq 1, \quad J_\zeta^{q_\zeta} P_\zeta = 0, \qquad d = \max_\zeta (q_\zeta - 1).$$

The q_ζ may be upper bounds; minimal orders give the weakest moment requirement. Write $Q = I - \sum P_\zeta$ and $E_< = EQ$. On the Riesz blocks,

$$E^n P_\zeta = \zeta^n \sum_{t=0}^{q_\zeta - 1} \binom{n}{t} \zeta^{-t} J_\zeta^t P_\zeta \quad (n \geq 0). \tag{2}$$

There exist $0 < \rho < 1$ and $A \geq 1$ such that

$$\|E^n Q\| \leq A\rho^n \quad (n \geq 0), \qquad \|E_<^n\| \leq A\rho^n \quad (n \geq 0). \tag{3}$$

The latter includes $E_<^0 = I$; the former includes Q. Existence is a compact spectral fact, but usable numerical values require a separate infinite-operator certificate.

For $z \in \mathbb{R}$ define the matrix polynomial

$$B_\zeta(z) = \sum_{t=0}^{q_\zeta - 1} \binom{z}{t} \zeta^{-t} J_\zeta^t P_\zeta.$$

Set

$$H_\zeta = (d+1)^d \sum_{t=0}^{q_\zeta - 1} \frac{\left\| J_\zeta^t P_\zeta \right\|}{t!}, \qquad H = \sum_\zeta H_\zeta, \qquad K = A + H. \tag{4}$$

For $d = 0$ the factor is one. Since $|z - j| \leq (d+1)(1 + |z|)$ for $0 \leq j < t \leq d$, these constants give

$$\|B_\zeta(z)\| \leq H_\zeta (1 + |z|)^d, \qquad \|E^n\| \leq K(1+n)^d. \tag{5}$$

Trace-class multiplication and the vanishing trace of positive nilpotent powers also give, for $n \geq 2$,

$$\left| \operatorname{tr} E^n - \sum_\zeta m_\zeta \zeta^n \right| \leq A \, \|E_<\|_2^2 \, \rho^{n-2}. \tag{6}$$

This does not say that $\operatorname{tr} E^n$ is positive or bounded away from zero.

Let $\mathscr{W}_+, \mathscr{W}_-$ be countable indexed families of bounded operators with nonnegative integer lengths. Repeated equal operators with different indices remain separate summands. For a pair write

$$\ell = \ell(W_+) + \ell(W_-), \quad w = \|W_+\| \, \|W_-\|, \quad M_r = \sum_{W_+, W_-} w(1+\ell)^r, \quad \tau_r(L) = \sum_{\ell > L} w(1+\ell)^r.$$

For integer $s \geq 0$ let

$$F_s = \sum_{\ell \leq s} \operatorname{tr}(C W_+ E^{s-\ell} W_-). \tag{7}$$

The key geometric input is the *difference* limit

$$F_s - R \operatorname{tr} E^{s+a} \longrightarrow 0, \qquad a \in \mathbb{Z}_{\geq 0}, \quad s + a \geq 2. \tag{8}$$

We neither replace it by a quotient limit nor infer it from finite data.

3 Polynomial moments and explicit discarded tails

Theorem 3.1 (Jordan-adapted extraction)**.** *In the preceding setting assume only $M_d < \infty$. The polynomials*

$$P_\zeta(z) = \sum_{W_+, W_-} \zeta^{-\ell} \operatorname{tr}(C W_+ B_\zeta(z - \ell) W_-) \tag{9}$$

converge coefficientwise absolutely. For $s \geq 2$ and $L = \lfloor s/2 \rfloor$,

$$\left| F_s - \sum_\zeta \zeta^s P_\zeta(s) \right| \leq D_s, \quad D_s = \|C\|_1 \left[A\rho^{s-L} M_0 + (2^d K + 3^d H) \tau_d(L) \right]. \tag{10}$$

If (8) holds, then

$$P_\zeta(z) \equiv R m_\zeta \zeta^a, \qquad |R| \le \overline{R} := \|C\|_1 \, M_d \min_\zeta \frac{H_\zeta}{m_\zeta}. \tag{11}$$

Consequently the same infinite model has the finite-index bound

$$|F_s - R \operatorname{tr} E^{s+a}| \le \epsilon_s := D_s + \overline{R} A \, \|E_<\|_2^2 \, \rho^{s+a-2}. \tag{12}$$

Proof. Coefficients of $\binom{z-\ell}{t}$ are bounded by a constant depending only on t times $(1+\ell)^t$. Thus M_d dominates every coefficient series in (9). It also proves absolute convergence of (7), by (5) and $M_0 \le M_d$.

For $\ell \le L$, replace $E^{s-\ell}$ by its exact peripheral part. The trace-ideal inequality bounds the total error by $\|C\|_1 A \rho^{s-L} M_0$. For $L < \ell \le s$, the retained actual powers have norm at most $K(1+s)^d$. Since $1+s \le 2(1+\ell)$, their total is at most $\|C\|_1 \, 2^d K \tau_d(L)$. For all $\ell > L$, including $\ell > s$, the polynomial part is bounded by $H(1+|s-\ell|)^d \le 3^d H (1+\ell)^d$. Subtract the full polynomial sum and apply the triangle inequality to these two tails. This proves (10). Both terms tend to zero.

Combining this with (6) and (8) gives

$$\sum_\zeta \zeta^s (P_\zeta(s) - R m_\zeta \zeta^a) \to 0.$$

For completeness, a finite sum $\sum \zeta^s Q_\zeta(s)$ of distinct unit-circle frequencies cannot tend to zero unless every polynomial vanishes. Divide by the highest power s^j, multiply by $\bar{\eta}^s$, and take Cesaro means. Geometric sums annihilate all frequencies but η, leaving its leading coefficient; lower terms tend to zero. Repeat through the finitely many degrees. This proves the identity in (11). Evaluation at zero and (5) give its bound for each ζ. Finally subtract the peripheral bare trace from the now constant polynomial sum, using (6), to obtain (12). □

What is improved. The source lemma assumes exponential moments of both word families. The theorem needs only M_d, where d records actual peripheral Jordan growth, and supplies a tail modulus explicitly. If $d = 0$, ordinary absolute summability suffices, even when there is no exponential word moment. Jordan expansion, trace ideals and Cesaro uniqueness are classical; the increment is the weaker summability hypothesis with a discarded-tail ledger at this particular source interface.

Example 3.2 (Why the degree matters)**.** For $d \ge 1$, take a single $(d+1)$-dimensional upper Jordan block $E = I + J$, $C = e_{d+1,1}$, $W_{+,\ell} = \ell^{-d-1} I$ of length $\ell \ge 1$, and $W_- = I$ of length zero. Then $\operatorname{tr}(CJ^t) = 0$ for $t < d$ and $\operatorname{tr}(CJ^d) = 1$. Every length moment of order $r < d$ is finite, but the series for $P_1(0)$ contains $\ell^{-d-1}\binom{-\ell}{d}$, all of sign $(-1)^d$ and of magnitude asymptotic to $1/(d!\ell)$. It diverges. This demonstrates failure of universal absolute coefficient extraction with lower moments. It is *not* a counterexample satisfying (8) for finite R.

Uniformity is a separate condition. Uniform spectral constants and a uniform bound on M_d give a uniform bound on R, but not a uniform rate in (12). To obtain that rate one needs $\sup \tau_d(L) \to 0$, together with uniform complementary spectral data. One pair of length N and weight $(1+N)^{-d}$ has $M_d = 1$ but $\tau_d(L) = 1$ whenever $N > L$. No finite collection of inspected words rules out this escape of weighted mass.

4 Anchored defect words with algebraic decay

The source word is anchored at the last selected defect: its length is not merely the number of selected defects. This distinction makes a useful explicit test possible. Let Δ_k be bounded operators, $k \ge 0$,

and for a finite nonempty set $S = \{k_1 < \cdots < k_j\}$ set

$$W(S) = E^{k_1}\Delta_{k_1}E^{k_2-k_1-1}\Delta_{k_2}\cdots E^{k_j-k_{j-1}-1}\Delta_{k_j}, \qquad \ell(S) = k_j + 1. \tag{13}$$

Use $W(\varnothing) = I$, $\ell(\varnothing) = 0$. Suppose $\|E^n\| \le A_1(1+n)^d$, $A_1 \ge 1$, and $\|\Delta_k\| \le \delta_k$. Define $b_k = A_1(1+k)^d\delta_k$.

Proposition 4.1 (Largest-index summation)**.** *For any real $r \ge 0$,*

$$\sum_S (1+\ell(S))^r \,\|W(S)\| \le 1 + \sum_{k\ge 0}(k+2)^r b_k \prod_{j<k}(1+b_j), \tag{14}$$

$$\sum_{\ell(S)>L} (1+\ell(S))^r \,\|W(S)\| \le e^B \sum_{k\ge L}(k+2)^r b_k, \quad B = \sum_{k\ge 0} b_k < \infty, \quad L \in \mathbb{Z}_{\ge 0}. \tag{15}$$

In particular, if $\delta_k \le c(1+k)^{-\beta}$ and $\beta > d + r + 1$, then, for $L \ge 1$,

$$\sum_{\ell(S)>L} (1+\ell(S))^r \,\|W(S)\| \le \frac{e^B A_1 c\, 2^r}{\beta - d - r - 1} L^{-(\beta-d-r-1)}, \quad B \le A_1 c\left(1 + \frac{1}{\beta - d - 1}\right). \tag{16}$$

Proof. Each gap preceding Δ_k has length at most k, so submultiplicativity gives $\|W(S)\| \le \prod_{k\in S} b_k$. Among subsets with largest index k, summing over all earlier selections gives exactly $b_k \prod_{j<k}(1+b_j)$. Their common weight is $(k+2)^r$. This proves (14), first for finite sets of available indices and then by monotone convergence of nonnegative majorants. For $\ell(S) > L$ the largest index is at least L, and $\prod(1+b_j) \le e^B$, proving (15). Finally $(k+2)^r \le 2^r(k+1)^r$ and $\sum_{k\ge L}(k+1)^{-u} \le \int_L^\infty t^{-u}\,dt$ for $u > 1$. The same integral comparison starting after the first term bounds B. □

For two end families, write $S_\pm = \sum_W (1+\ell(W))^d \,\|W\|$ and $T_\pm(L)$ for their corresponding weighted tails. The elementary inequality $1 + u + v \le (1+u)(1+v)$ gives

$$M_d \le S_+ S_-, \qquad \tau_d(L) \le T_+(L/2)S_- + S_+T_-(L/2). \tag{17}$$

The real threshold in the right side means the exact strict inequality in the definition of the tail; integer floors may be taken accordingly. Thus $\beta > 2d + 1$ is a simple sufficient algebraic decay condition for Theorem 3.1. It is not claimed necessary.

Example 4.2 (An explicit nonexponential regime)**.** Let $d = 1$, $A_1 = 2$ and $\delta_k \le (1+k)^{-4}/100$ at each end. Then $b_k \le (1+k)^{-3}/50$, $B \le 3/100$, and for integer $L \ge 1$

$$T_\pm(L) \le \frac{e^{3/100}}{25L}, \qquad S_\pm \le 1 + \frac{2e^{3/100}}{25}.$$

Here the second bound uses $\sum_{k\ge 0}(k+1)^{-2} \le 2$. Equations (17) and (10) give a certified $O(s^{-1})$ word-tail error plus complementary spectral decay. The source physical words have exponential defect decay and already meet stronger hypotheses. This example extends the abstract interface; it does not construct a new honeycomb gas with these algebraic defects.

5 Cancellation-safe finite trace windows

Let $L \ge 1$ and $N \ge 0$ be integers with $N + a \ge 2$. For $\mathcal{I} = \{N, \ldots, N + L - 1\}$, put $T_s = \operatorname{tr} E^{s+a}$, and use the Euclidean norm on this finite complex sample space. Set $\langle T, F\rangle = \sum \overline{T_s}F_s$. The samples must be from the same operator and central insertion.

Theorem 5.1 (Window and data error). *If $T \neq 0$ and $|F_s - RT_s| \le \epsilon_s$, then*

$$\widehat{R} = \frac{\langle T, F\rangle}{\|T\|_2^2}, \qquad |\widehat{R} - R| \le \frac{\|\epsilon\|_2}{\|T\|_2}. \tag{18}$$

Among complex linear estimators $\sum c_s F_s$ that recover every R when $F = RT$, this has the smallest worst-case error for an arbitrary Euclidean residual ball.

For perturbed data $\widetilde{T}, \widetilde{F}$, suppose $\left\|T - \widetilde{T}\right\|_2 \le \delta$ and $\left\|\widetilde{F} - RT\right\|_2 \le e$. With $\widetilde{R} = \langle \widetilde{T}, \widetilde{F}\rangle / \left\|\widetilde{T}\right\|_2^2$, the following two alternatives hold:

$$|\widetilde{R} - R| \le \frac{e + \overline{R}\delta}{\left\|\widetilde{T}\right\|_2} \qquad \text{if } |R| \le \overline{R},\ \left\|\widetilde{T}\right\|_2 > 0, \tag{19}$$

$$|\widetilde{R} - R| \le \frac{e + |\widetilde{R}|\delta}{\left\|\widetilde{T}\right\|_2 - \delta} \qquad \text{if } \left\|\widetilde{T}\right\|_2 > \delta. \tag{20}$$

Here one may use $e = \|\epsilon\|_2 + \nu$ if $\left\|\widetilde{F} - F\right\|_2 \le \nu$ and (12) is available.

Proof. Subtract RT in the numerator and apply Cauchy–Schwarz. An unbiased coefficient vector satisfies $\sum c_s T_s = 1$, hence $\|c\|_2 \|T\|_2 \ge 1$. Equality is attained by $c_s = \overline{T_s} / \|T\|_2^2$; the worst residual is proportional to $\bar{c}$, so the minimax statement is exact for this specified ball. For perturbed data, projection gives

$$|\widetilde{R} - R| \left\|\widetilde{T}\right\|_2 \le \left\|\widetilde{F} - R\widetilde{T}\right\|_2 \le e + |R|\delta.$$

The prior bound gives (19). Substituting $|R| \le |\widetilde{R}| + |\widetilde{R} - R|$ and absorbing the last term gives (20). No assumption on a sign of any sample is used. □

5.1 A lower bound for the whole window

Write $p = |\mathcal{P}|$, let $V_L = (\zeta^j)_{0 \le j < L, \zeta \in \mathcal{P}}$, and let e^{tr} be the complementary-trace remainder from (6) on this window. Then

$$\|T\|_2 \ge \sigma_{\min}(V_L) \left(\sum_{\zeta} m_\zeta^2\right)^{1/2} - \left\|e^{\mathrm{tr}}\right\|_2 \qquad (L \ge p). \tag{21}$$

Indeed, the peripheral coefficient vector is $(m_\zeta \zeta^{N+a})_\zeta$ and has the displayed norm, independently of starting phase. Distinct frequencies make the square Vandermonde V_p invertible. Thus p consecutive samples remove the possibility of complete peripheral cancellation, even for nonsemisimple operators. They do not guarantee a large conditioning constant when frequencies are close.

A coarser bound avoiding a singular-value calculation follows from the Gram matrix. Its diagonal is L and its off-diagonal entries are geometric sums. Define

$$B = \max_{\zeta \in \mathcal{P}} \sum_{\eta \neq \zeta} \frac{2}{|1 - \bar{\zeta}\eta|}.$$

For any vector v, bounding each cross term by $|v_\zeta|^2 + |v_\eta|^2$ gives $v^* V_L^* V_L v \ge (L - B)\|v\|_2^2$. Consequently $\sigma_{\min}(V_L) \ge \sqrt{L - B}$ whenever $L > B$. The exact Gram entries or $\min\{L, 2/|1 - \bar{\zeta}\eta|\}$ can sharpen this lower bound. Certified eigenvalue separation and complementary trace tails remain inputs, not outputs of a finite matrix fit.

Example 5.2 (A cancelled trace with Jordan blocks)**.** Take $E = J_2(1) \oplus J_2(-1)$, with superdiagonal one in each block. For every positive n,

$$\operatorname{tr} E^n = 2 + 2(-1)^n.$$

Every odd sample vanishes, but every pair of consecutive samples has Euclidean norm four. For the empty-word family, let $a = 0$ and $C = RI$. Then $F_s = R \operatorname{tr} E^s$, and a two-sample residual of norm e gives error at most $e/4$, attained by a residual proportional to T. For general C, the two block traces may additionally contain $s \operatorname{tr}(C_+ J)$ and $s(-1)^{s-1} \operatorname{tr}(C_- J)$. The identification limit forces those polynomial coefficients to vanish; bare trace cancellation alone would not remove them. This is why the Jordan moment calculation and the window bound address different issues.

Finite work versus conditional information. Equations (18)–(20) are finite computations once intervals for samples, spectral constants and tails are supplied. An unquantified source $o(1)$ is not an algorithm to supply those intervals. Likewise, truncating the infinite Bargmann memory or the charge coordinate does not automatically certify (3). The theorem does not claim a numerical cylinder amplitude has been computed.

6 The marked two-gap extension

The marked source has different limiting exterior operators E_+ and E_-, both Hilbert–Schmidt of radius one. Give each its own Riesz data as above, with maximum Jordan degrees d_+, d_-, and put $h = d_+ + d_-$. Four word families have lengths ℓ_i and norms with product w; write $\ell_+ = \ell_1 + \ell_2$, $\ell_- = \ell_3 + \ell_4$ and $\ell = \ell_+ + \ell_-$. At fixed integer $L \geq 0$, $g = 2N - L$, define

$$F_{N,L} = \sum_{\ell_+, \ell_- \leq g} \operatorname{tr}(C W_1 E_+^{g-\ell_+} W_2 W_3 E_-^{g-\ell_-} W_4). \tag{22}$$

The required source identification is $F_{N,L} - R \operatorname{tr} E_+^{4N} \to 0$ as $N \to \infty$ at this fixed L.

Theorem 6.1 (Polynomial moments for two gaps)**.** *Replace the source four exponential word bounds by*

$$M_h^{(4)} = \sum_{W_1, W_2, W_3, W_4} w(1+\ell)^h < \infty. \tag{23}$$

For $\zeta \in \mathcal{P}_+$ and $\eta \in \mathcal{P}_-$ define

$$P_{\zeta,\eta}(z) = \sum_W \zeta^{-\ell_+} \eta^{-\ell_-} \operatorname{tr}(C W_1 B_\zeta^+(z - \ell_+) W_2 W_3 B_\eta^-(z - \ell_-) W_4).$$

These are coefficientwise absolutely convergent polynomials of degree at most h. For every frequency β in the union of $\{\alpha^4 : \alpha \in \mathcal{P}_+\}$ and $\{(\zeta\eta)^2 : \zeta \in \mathcal{P}_+, \eta \in \mathcal{P}_-\}$,

$$R \sum_{\alpha^4 = \beta} m_\alpha^+ = \sum_{(\zeta\eta)^2 = \beta} (\zeta\eta)^{-L} P_{\zeta,\eta}(-L). \tag{24}$$

In particular $|R| \leq K_*(1+L)^h \|C\|_1 M_h^{(4)}$, where $K_* < \infty$ depends only on the two sets of peripheral data, not on L, C, R.

Proof. The product of the two binomial polynomials has degree at most h, and each coefficient is bounded by a fixed constant times $(1+\ell)^h$. This proves coefficientwise convergence. On the terms with both $\ell_\pm \leq g/2$, replacing the two powers by their peripheral expansions has total error bounded

by $C_* \|C\|_1 (1+g)^h \rho^{g/2} M_h^{(4)}$, after increasing $\rho < 1$ and the constant to cover both complements. This follows by subtracting one factor at a time and applying the two polynomial power bounds.

On every remaining term, $\ell > g/2$. The product of actual power norms, when both powers exist, is at most a constant times $(1+g)^h$. The product of the two peripheral polynomial norms, including terms outside the retained region, is bounded by a constant times $(1+g+\ell)^h$. Each is at most a constant times $(1+\ell)^h$ on this tail. Its total tends to zero by (23). Hence

$$F_{N,L} = \sum_{\zeta,\eta} (\zeta\eta)^{2N-L} P_{\zeta,\eta}(2N-L) + o(1).$$

Combine this with the bare trace expansion for E_+^{4N} and group *equal frequencies in N before* Cesaro extraction. This gives the polynomial identity

$$\sum_{(\zeta\eta)^2=\beta} (\zeta\eta)^{-L} P_{\zeta,\eta}(2z-L) \equiv R \sum_{\alpha^4=\beta} m_\alpha^+.$$

Its value at zero is (24). Choose $\beta = \alpha_0^4$; the multiplier of R is a positive integer. At $z = -L$, the norm bounds give $|P_{\zeta,\eta}(-L)| \le C_* \|C\|_1 (1+L)^h M_h^{(4)}$. Summing the finite group proves the bound. □

If both peripheral spectra contain $1, -1$, all products $(\zeta\eta)^2$ and all fourth powers equal one. There are four pairs on the right, not one. Extracting an individual product coefficient would be false in general. In the source, the marked central trace estimate uses monotone common-charge jumps only *after* omission of suppressed particles. The mixed spectral radius and buffers are separate from the preceding lemma. The total moment condition here is convenient and sufficient; no minimal anisotropic moment condition is claimed.

7 Physical certificates and the original Ram comparison

7.1 Finite moment coordinates do not alter the amplitude

The freshly read original finite-moment proof [4] uses an unisolvent finite polynomial bank and its compatible linear functionals. It supports exact transport of a finite polynomial identity, not automatic control of a continuum operator or of its omitted memory states. In the present source this applies to a *cleared* finite selected-vector identity with its proven degree and denominator. It cannot turn generic individually singular spin-sum terms into evaluations at a collision: the combined polynomial must first be continued there.

For clarity, if $J : \mathbb{C}^q \to \mathcal{H}$ is injective and $G = J^*J > 0$, the correct coefficient norm is $\|c\|_G^2 = c^*Gc$. A coordinate operator B acting on this subspace has norm

$$\|B\|_G = \left\| G^{1/2} B G^{-1/2} \right\|_2 . \tag{25}$$

This follows by the isometry $c \mapsto JG^{-1/2}c$. Likewise a physical functional with coefficient row λ^* has dual norm $\sqrt{\lambda^* G^{-1} \lambda}$, since Cauchy–Schwarz after the same change of variables is sharp. The original physical-metric chapter [5] proves this synthesis/Gram distinction. A nonunitary fusion or polynomial-basis change does not preserve Euclidean singular values or the constants in (4).

If $P = JG^{-1}J^*$ and $B = G^{-1}J^*EJ$, the full residual on a synthesized vector is

$$EJc - JBc = (I - P)EJc.$$

A zero compressed residual says nothing about this orthogonal component. For a full block decomposition $E = \left(\begin{smallmatrix} B & U \\ V & D \end{smallmatrix}\right)$, a putative resolvent certificate at z must also control $z - D$ and the Schur term $U(z-D)^{-1}V$. It is not supplied by a small residual of B alone. The fresh original spectral-limits chapter [6] contains explicit finite-section pollution and graph-domain cautions. We use that boundary, not a claim that its self-adjoint examples prove stability of this nonnormal signed operator.

7.2 Residual accounting at the source interface

Suppose finite computation supplies $\widetilde{F}_s, \widetilde{T}_s$. There are at least four distinct budgets:

1. arithmetic and finite physical-coordinate errors in the computed traces;
2. omitted anchored words, bounded in the appropriate physical norm by (15) and (17);
3. complementary spectral and infinite charge/memory errors, which require a whole-operator certificate such as (3);
4. the geometric identification of the limiting scalar with the source finite-list regulator or marked polygon amplitude.

Theorem 5.1 combines the first three when valid bounds are given. It does not prove the fourth. Nor does an exact finite polynomial rank test establish uniformity as cylinder size, period or contour displacement changes. The relevant original moment proof is a finite duality argument; it supplies no such uniformity by itself.

8 What remains in the geometric conclusion

For the balanced cylinder the source scales the central operator at $L = \lfloor \log m/(\kappa_1 b) \rfloor$, $\kappa_1 = \pi/3$. Its exterior radius is $r_E = e^{-\pi b/16}$, while its restricted interior radius is $u(\theta) = e^{-4\pi b(\theta/\pi)^2/9}$ for real $|\theta| < \pi/2$. The central comparison therefore contributes $3/8 - 8\theta^2/(3\pi^2)$ to the logarithmic power. The independent homogeneous scalar normalization contributes $5/24$, yielding $1/6 - 8\theta^2/(3\pi^2)$, or $1/6 - 2\Theta^2/(3\pi^2)$ with $\Theta = 2\theta$. These normalizations explain why a coordinate improvement in a trace bound does not improve the cylinder exponent.

The source obtains the $5/24$ scalar from a confluent Pfaffian ratio:

$$N(1,\dots,1) = \left(\frac{2(2+\sqrt{2})}{3}\right)^{m(2m-1)} \left(\frac{2}{3}\right)^{m} \det(P_m e^{f} P_m|_{\operatorname{ran} P_m}),$$

where P_m is the even-polynomial projection in $L^2(2u/\sinh(\pi u)\,du)$. Its logarithmic determinant has exponent $5/48$; squaring supplies $5/24$. The proof uses two kernel peaks at $v = u$ and $v = -u$. Evenness of f is needed at the second peak, and a finite Gram determinant alone is not the logarithmic asymptotic. The complete cached Pfaffian module was read, but its external cited orthogonal-polynomial literature was not independently audited.

For marked intervals, the source additionally needs the trace-class central bound involving the restricted common-charge radius $u_c = e^{-\pi b/4}$ and mixed radius

$$r_X^{+} = \exp\left[b\left(-\frac{5\pi}{96} + \frac{1}{6\sqrt{3}}\right)\right].$$

After buffers, small spectral slack, and scalar normalization, the exponents are

$$-\frac{5}{24} + \frac{2\log(u_c/r_E)}{\kappa_1 b} = -\frac{4}{3}, \qquad -\frac{5}{24} + \frac{2\log(r_X^{+}/r_E)}{\kappa_1 b} = \frac{1}{\pi\sqrt{3}} - \frac{7}{48} < \frac{1}{3}.$$

Together with the source identification this gives its $n^{o(1)} m^{-4/3} (n/m)^{1/3}$ bound for $n \geq m$ sufficiently large, with the fixed count-discrepancy and height-proportion restrictions. Theorem 6.1 merely replaces one moment assumption in this chain. The mixed spectral comparison and planar injection are not consequences of it.

In particular, period tends to infinity with finite lists fixed before large-list uniform estimates are used; physical spread-one heights are approached through the regular finite marked matrix element.

The source does not require a raw Gaussian field to remain analytic across collision. Pre-collision Gaussian integration, meromorphic contour deformation, and post-collision stability are ordered operations. An estimate proved on fixed generic heights does not authorize exchanging all these limits.

8.1 Benefits, costs, and feasible extensions

The principal benefit is a weaker, explicitly degree-dependent tail requirement at the exact signed-trace interface. Algebraic defects can now be handled when their weighted moments are finite. The window estimator tolerates zeros of individual bare traces, and its a posteriori form (20) does not require a separate prior bound on R. The two-gap extension preserves frequency collisions instead of selecting an arbitrary peripheral pair.

There are costs. Riesz projections can have large norms, Jordan orders may be hard to certify, nearly coincident frequencies give ill-conditioned windows, and the safe tail constants are not optimized. No running-time speedup is measured. The source already has exponential physical word decay, so the improvement enlarges the abstract admissible perturbation class rather than strengthening its stated polygon exponent.

Three plausible next steps have explicit missing inputs. First, a computer-assisted cylinder amplitude would need certified complementary resolvents and a quantitative physical word-tail modulus, not just matrix eigenvalues. Second, a sharper anisotropic two-gap moment theorem could track different lengths against individual nilpotent degrees, but must control the region where one gap is short while the other grows. Third, higher-degree window fitting could estimate the full exponential polynomial before imposing the identification limit. This requires confluent Vandermonde conditioning and certified degree/spectral data; we claim no such validated numerical reconstruction here.

9 Read scope, checks, and assessment

The owned ledger records pre-read hashes before fresh displays, confirmed literal line ranges and their unread complements, and matching post-read hashes. It includes all 21 public text endpoints, with complete reads of the local, finite, contour, stability, compact propagation, trace, Pfaffian, cylinder-exponent, localization, marked, and shifted modules, plus the introduction, driver, bibliography, metadata and figure source code. Longer boundary, bosonization, marked-gas and mixed modules were read in selected proof intervals recorded below. Figure source code was read; this is not a claim to have inspected the source figure renderings.

Partially read source	Literal intervals; all complements remain uninspected
Boundary, 1147 lines	1–148, 926–1147
Bosonization, 1218 lines	1–165, 882–1218
Marked gas, 567 lines	1–174, 477–567
Mixed propagation, 884 lines	1–114, 372–884

Thus the boundary positive-integral asymptotics, the middle oscillator and collision calculations, part of the marked gas reorganization, and the interior one-species mixed-radius proof remain imported rather than independently verified here. Mapping their place in the architecture is not equivalent to reading or certifying them.

Fresh original private reads were TR250 lines 145–270 and 455–594, the physical-metric chapter lines 1–210, and the spectral-limits chapter lines 1–242. They support the finite-moment and physical-norm comparison, not the honeycomb topology or gas construction. Included files and the rest of the private corpus were not thereby read. All private material remained local.

The owned executable checks use exact rational arithmetic on generalized binomials, Jordan powers, finite anchored-word sums, signed trace windows, data perturbations, and grouped two-gap fixtures: 5,146 parameter cases and 9,726 assertions passed. Group counts and exact script hashes are recorded in the accompanying test receipt. These are independent finite sanity checks of the displayed local algebra, not replay of source or parent checkers and not validation of an infinite gas or planar exponent. The freeze receipt binds the exact final report, tests, read ledger, page images and read-only input endpoints.

Applicability remains L3. The C3 assessment is for the polynomial-moment weakening with explicit tails, its anchored algebraic-defect criterion, and the quantitative cancellation-safe extraction interface. Classical spectral, trace-ideal, least-squares and Vandermonde tools are credited. No global proof certification, new self-avoiding-walk law, numerical amplitude, or stronger polygon exponent is claimed. No source correction is asserted on the performed scope; the report adds conditional local results while retaining the source's valid distinctions.

References

[1] OpenAI mathematics preprint repository. *Signed cylinder propagation and marked polygons on the honeycomb lattice*. September 26, 2026. Captured source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, authenticated unchanged against the captured October 8 tree `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. This is byte provenance, not a live-branch or proof-validity assertion.

[2] H. Duminil-Copin and S. Smirnov. *The connective constant of the honeycomb lattice equals* $\sqrt{2+\sqrt{2}}$. Annals of Mathematics 175 (2012), 1653–1665. https://annals.math.princeton.edu/2012/175-3/p14. Fresh primary abstract and metadata read; full external proof not audited.

[3] D. Krachun and C. Panagiotis. *Quantitative sub-ballisticity of self-avoiding walk on the hexagonal lattice*. https://arxiv.org/abs/2310.17299, v1, October 26, 2023. Source bibliography records Annals of Probability 54 (2026), 1109–1125, doi:10.1214/24-AOP1730. Fresh arXiv abstract and metadata read; publisher full text was not obtained.

[4] M. Subbarao. Original private Ram Foundation manuscript, `TR-rev-1-old-tr250.tex`, lines 145–270 and 455–594. Fresh local finite-moment, compatibility and finite-quadrature proof passages; no whole-file or whole-corpus read claimed.

[5] M. Subbarao. Original private Ram Calculus Books, volume 21, `ch01d-physical-metrics.tex`, lines 1–210. Physical synthesis, Gram norms and omitted residuals.

[6] M. Subbarao. Original private Ram Calculus Books, volume 21, `ch01f-spectral-limits.tex`, lines 1–242. Finite-section pollution, graph domains and stability boundaries.